rezki ::= engineer | educator

Rezki Boureghdad

Back-end & Data Architect · Computer Science Educator

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PROFILE

Software engineer focused on back-end and data architecture, and a computer-science educator. At Solrus, an Algiers software studio, I design the data models, APIs and integrations behind client production platforms. I also teach at Université M'Hamed Bougara, where I have covered 12 subjects, from algorithms and databases to compilation and semantics, at Bachelor's and Master's level. I take the same approach in both roles: get the structure right, make it measurable, and explain it clearly.

EXPERIENCE

Data Specialist, Structure & Architecture

2025 – Present

Solrus — software studio for AI, cloud and secure systems · Algiers, remote-first

- Own back-end and data architecture for client platforms: domain modelling, relational schemas and migrations, API contracts, and third-party integrations.
- Build event-driven and async Python services (FastAPI, SQLAlchemy 2, Kafka, Redis, Celery) and tune them for latency and reliability.
- Security in the design from the start: role-based access, 2FA, audit trails, data-residency compliance, hardening before audits.
- Test-first delivery with pytest against real PostgreSQL and end-to-end Playwright suites. Work closely with front-end, AI and product teammates.

> Most client projects are under NDA. Names and details are not disclosed.

Freelance Full-Stack Developer (auto-entrepreneur)

May 2021 – Present

Independent · web, mobile and AI products

- Run projects end to end for local businesses and community organisations: scoping, proposal, clickable prototype, build and deployment.
- Member of the team behind **Zakat.business**, an award-winning non-profit startup project.

Teaching Assistant — Computer Science

Nov 2024 – Present

Université M'Hamed Bougara, Boumerdès

- 12 subjects taught so far to Software Engineering and LMD students, Bachelor's and Master's level.
- Prepare and run tutorials and labs, write exercises and assessments, and help students connect theory to hands-on practice.
- Adjust my teaching for each audience, from first programming courses to formal semantics for Master's students.

Development Mentor

Oct 2024 – Sep 2026

DevUp Club, Université M'Hamed Bougara

- Designed and ran practical software-engineering workshops for students, with hands-on material that participants applied to real projects.

EDUCATION

MSc Software Engineering & Data Science
UMBB · Jan 2024

BSc Computer Science
UMBB · Jan 2022

SUBJECTS TAUGHT

Algorithms & Data Structures

Databases

Programming Languages (intro → advanced)

Compilation

Semantics

+ 7 other CS modules

CERTIFICATIONS

NVIDIA
Fundamentals of Deep Learning · Intro to Machine Learning

Neo4j
Graph Data Science · Fundamentals · Graph Data Modeling

EF SET English
73/100 — C2 Proficient

LANGUAGES

Arabic
Native

English
C2 (EF SET)

French
Professional working

SELECTED PROJECTS

Public, open and personal work only.

Forsa / Aamal ATS — recruitment platform

Solrus public case study

- Connects companies, independent recruiters and job seekers in Algeria. Semantic job matching with sentence-transformer embeddings and location search with PostGIS.
- WebSocket chat, community forum, FCM push notifications, SlickPay subscriptions, JWT/Argon2 auth. Flutter mobile app and admin panel.

FastAPI · PostgreSQL/PostGIS · Flutter · Docker

e-motion — back-office for cash-on-delivery shops

Own product

- Orders, stock, couriers and expenses, with real profit per order. Weighted-average purchase cost, stock that can never go below zero, alerts for customers who often cancel or return, and an audit log.
- Excel import, packing slips, English/French interface. Runs on free hosting tiers.

Next.js · TypeScript · Supabase · Vercel

Ahl Quran — multi-tenant school manager

Community

- Study circles, attendance, student progress, messaging and reports, separated by organisation. Web and mobile apps.

FastAPI · PostgreSQL · Flutter/GetX · Docker

7alkaty — mosque course & imam-following app

Mobile

- REST API documented with Swagger, role-based access, scheduled push notifications, media storage, and a React Native client.

Node/Express · MongoDB · Expo

Forest-fire detection & reporting

Master's thesis

- Co-designed a three-part system: a citizen reporting app, a data platform and alerting. Detection models built with YOLO and Faster R-CNN.

Python · YOLO · R-CNN · Flutter · Firebase

ISO 27001 skill for Claude Code

Tooling

- Checks a codebase against ISO/IEC 27001:2022, plans the rollout step by step, and checks every edit with a scanner hook.

Python · Security compliance · AI tooling

SKILLS

Back-end
Python, FastAPI, Django/DRF, SQLAlchemy 2 (async), Alembic, Pydantic, Celery, Node/Express

Data
PostgreSQL/PostGIS, Redis, Kafka, MongoDB, Neo4j, Elasticsearch, Supabase

AI / ML
RAG, LangGraph, FAISS/Chroma, embeddings & reranking, LLM APIs, YOLO/R-CNN

Front & mobile
TypeScript, Next.js, React, Tailwind, Flutter/Dart, React Native

Ops & quality
Docker, nginx, GitHub Actions, Vercel, Grafana/Loki, pytest, Playwright

Security
JWT, Argon2, OAuth2, TOTP 2FA, ISO 27001, data residency

Also
Java, C++, SQL; contract-first API design, decision records, AI-assisted development

CS FOUNDATIONS

Algorithms & complexity

Compilers & formal languages

Formal methods

Deep learning

Programming-language semantics

Database theory & graph databases